Polygenic Risk Scores

BIOS 25328 · Lecture 7

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2026-04-13

Learning Objectives

In this lecture we will learn about **Polygenic Risk Scores (PRS)** — how they are constructed, evaluated, and applied in research and clinical settings.

- Explain the polygenic additive model and its role in PRS calculation
- Describe the statistical challenges in estimating PRS and methods to address them
- Evaluate the clinical utility and limitations of PRS, including ancestry-related biases
- Discuss the ethical considerations surrounding the use of PRS in embryo screening
- Compare and contrast different approaches for calculating and improving PRS

The Polygenic Additive Model

Polygenic Additive Model

$$Y = \sum_{k}^{M} X_k \cdot \beta_k + \epsilon$$

. . .

Can also include covariates (sex, age, etc.):

$$Y = \text{covariates} + \sum_{k}^{M} X_k \cdot \beta_k + \epsilon$$

The most common model used to study the genetics of complex diseases today is the polygenic additive model. X_k represents the genotype at position k, _k is the effect size of SNP k. is the error term, typically representing the environmental component, but in practice it

accommodates everything that doesn't fit in the summation. One can add covariates such as sex and age to make the model more general.

Estimating Parameters of the Polygenic Additive Model

Can we fit all SNPs at the same time?

$$Y = \mu + a \cdot \text{age} + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_{1,000,000} X_{1,000,000}$$

Why can't we estimate betas by least squares (ordinary linear regression)?

. . .

Too many parameters and too few observations

We may wonder why we don't fit all the SNPs at the same time. The problem is that there are too many parameters and not enough data points. We typically have millions of SNPs and only thousands of individuals. Even with growing sample sizes, the estimates would overfit the data and not generalize to new individuals.

Rule of Thumb

10 data points per parameter

In a pinch, at least **5 data points per parameter**

Solution: Reduce Parameters via Random Effects

Assume $\beta_k \sim N(0, \sigma_\beta^2)$ and estimate just the variance σ_β^2

$$\text{From millions of } \beta_k \text{ parameters} \longrightarrow 1 \text{ variance parameter}$$

* This is one form of **regularization** — more on this later

Instead of fitting millions of s as fixed effects, we can consider them random and estimate their distribution. Assuming is normally distributed with mean 0 and variance ^{2}, we only need to estimate ^{2}. So from millions of parameters we are down to 1.

Mixed Effects Modeling

$$Y = \text{fixed effects} + \text{random effects} + \text{noise}$$

$$= \text{fixed effects} + \sum_k \beta_k X_k + \epsilon$$

. . .

β_k's are random

$$\beta_k \sim N(0, \sigma_\beta^2)$$

. . .

** this is one form of Regularization, more on this later

Connection to EMMAX

Recall the mixed effects approach used to correct for population structure:

$$Y = X_{\text{test}} \cdot \beta_{\text{test}} + u + \epsilon, \quad u \sim N(0, \sigma^2 \cdot \mathbf{K})$$

The random effect u is equivalent to the **sum of all SNP effects**:

$$Y = X_{\text{test}} \cdot \beta_{\text{test}} + \sum_k X_k \beta_k + \epsilon$$

To demonstrate that EMMAX's random effect u and the sum of all SNP effects are the same, we show they have the same covariance matrix — the genetic relatedness matrix. Calculate E[uu'] and you find it equals ²K, where K is the genetic relatedness matrix. QED.

Review: Norms of Vectors

L2 Norm of a Vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_M \end{bmatrix} \qquad \|X\|_2 = \sqrt{\sum_k x_k^2}$$

Our familiar Euclidean distance of a vector.

In a GWAS We Minimize the L2 Norm of the Error

In a GWAS we find one SNP at a time:

$$\mathbf{Y} = \mu + a \cdot \mathbf{age} + \beta_1 \cdot \mathbf{X} + \epsilon$$

Find μ, a, β_1 that minimizes:

$$\|\mathbf{Y} - \mu - a \cdot \mathbf{age} - \beta_1 \cdot \mathbf{X}\|_2^2$$

$$= (y_1 - \mu - a \cdot \text{age}_1 - \beta_1 x_1)^2 + (y_2 - \mu - a \cdot \text{age}_2 - \beta_1 x_2)^2 + \cdots$$

In traditional linear regression, we find estimates of , a, and that minimize the L2 norm of the error term.

Which One Is the L1 Norm?

Option 1:

$$= (y_1 - \mu - a \cdot \text{age}_1 - \beta_1 x_1)^1 + (y_2 - \mu - a \cdot \text{age}_2 - \beta_1 x_2)^1 + \cdots$$

Option 2:

$$= |y_1 - \mu - a \cdot \text{age}_1 - \beta_1 x_1| + |y_2 - \mu - a \cdot \text{age}_2 - \beta_1 x_2| + \cdots$$

Which definition makes more sense as the L1 norm — and why? Option 2 is correct. The L1 norm uses absolute values to ensure all terms are non-negative, making it a proper distance measure.

Prediction of Complex Traits

Prediction of complex traits can help us better tailor treatment of patients.

Simple Polygenic Risk Score

$$Y = \sum_{k=1}^{M} \hat{\beta}_k^{\text{GWAS}} X_k$$

Just use GWAS effect sizes — the seemingly naive approach yields better prediction than expected.

ISC (2009). Common polygenic variation contributes to risk of schizophrenia and bipolar disorder. *Nature*

Polygenic risk scores are simple to calculate: take the estimated effect sizes from the GWAS (output from plink, for example), multiply by each person's genotype, and sum. This seemingly naive approach yields much better prediction than one could expect.

Tricks to Deal with Too Many Parameters

1. **Model s as random effects** — estimate variance σ_β^2 instead of each β_k
2. **Use GWAS effects** — fit one SNP at a time, let absorb the rest
3. **Penalized likelihood / regularization** — penalize the L1 or L2 norm of the parameter vector:

$$\left\| Y - \sum_k X_k \beta_k \right\|_2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2$$

These three approaches trade off between complexity and interpretability. Regularization restricts the values parameters can take, preventing overfitting by penalizing large effect sizes.

BLUP / Ridge Regression

Best Linear Unbiased Prediction (BLUP) / Ridge

$$Y = \sum_{k=1}^{M} \hat{\beta}_k^{\text{Ridge}} X_k$$

Penalized regression $\longrightarrow$ **Ridge**

$$\left\| Y - \sum_k X_k \beta_k \right\|_2 + \lambda_2 \|\beta_2\|_2$$

Ridge regression (BLUP) minimizes the prediction error with a penalty on the L2 norm of the effect size vector. It tries to minimize the mean square error while keeping the length of the effect size vector small.

LASSO and Elastic Net

LASSO/Elastic Net Prediction

$$Y = \sum_{k=1}^{M} \hat{\beta}_k^{\text{E-N}} X_k$$

Penalized regression $\longrightarrow$ **LASSO** and **Elastic Net**

$$\left\| Y - \sum_k X_k \beta_k \right\|_2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta_2\|_2$$

LASSO penalizes the sum of absolute values of the effect sizes (L1 norm), yielding sparse models with a few large-effect SNPs. Elastic net mixes L1 and L2 norms, yielding less sparse models, though not quite polygenic ones.

Whole-Genome Prediction: BSLMM

$$Y = \sum_{k=1}^{M} \beta_k^L X_k + \sum_{k=1}^{M} \beta_k^S X_k + \epsilon$$

$$\beta_k^L \sim N(0, \sigma_L^2) \quad \text{(polygenic: many small effects)}$$
$$\beta_k^S \sim N(0, \sigma_S^2) \quad \text{(sparse: few large effects)}$$

Zhou, Carbonetto & Stephens (2013). Polygenic modeling with Bayesian sparse linear mixed models. *PLOS Genetics*

BSLMM models genetic effects as a mixture of two normals: a small-variance component capturing the polygenic background, and a large-variance component capturing sparse large effects. This flexibility allows appropriate sparsity to be learned from data. Other whole-genome approaches include multiBLUP and OmicKriging.

Advantages of Simple Polygenic Scores

Main advantage: easy to obtain and scalable

- GWAS summary statistics are publicly available
- No individual-level data required

vs. multivariate approaches (ridge, elastic net, BSLMM):

- Require individual-level data
- Although some fine-mapping methods allow inferring multivariate results from summary statistics

Improving PRS with LD Information

Methods that incorporate LD structure into PRS:

Method	Reference
PRSice (pruning + thresholding)	Choi & O'Reilly
Lasso-sum	Mak et al. 2017
LDPred	Vilhjálmsson et al. 2015
RSS	Zhu & Stephens 2017
S-BayesR	Lloyd-Jones et al. 2019
PRS-CS	Ge et al. 2019

RSS and S-BayesR are likelihood-based methods with different priors on s.

PRSice implements simple PRS calculation with calibration of p-value thresholds and pruning of correlated SNPs. The other methods infer multivariate regression results using summary statistics and reference LD information.

Importance of Good LD Reference Data

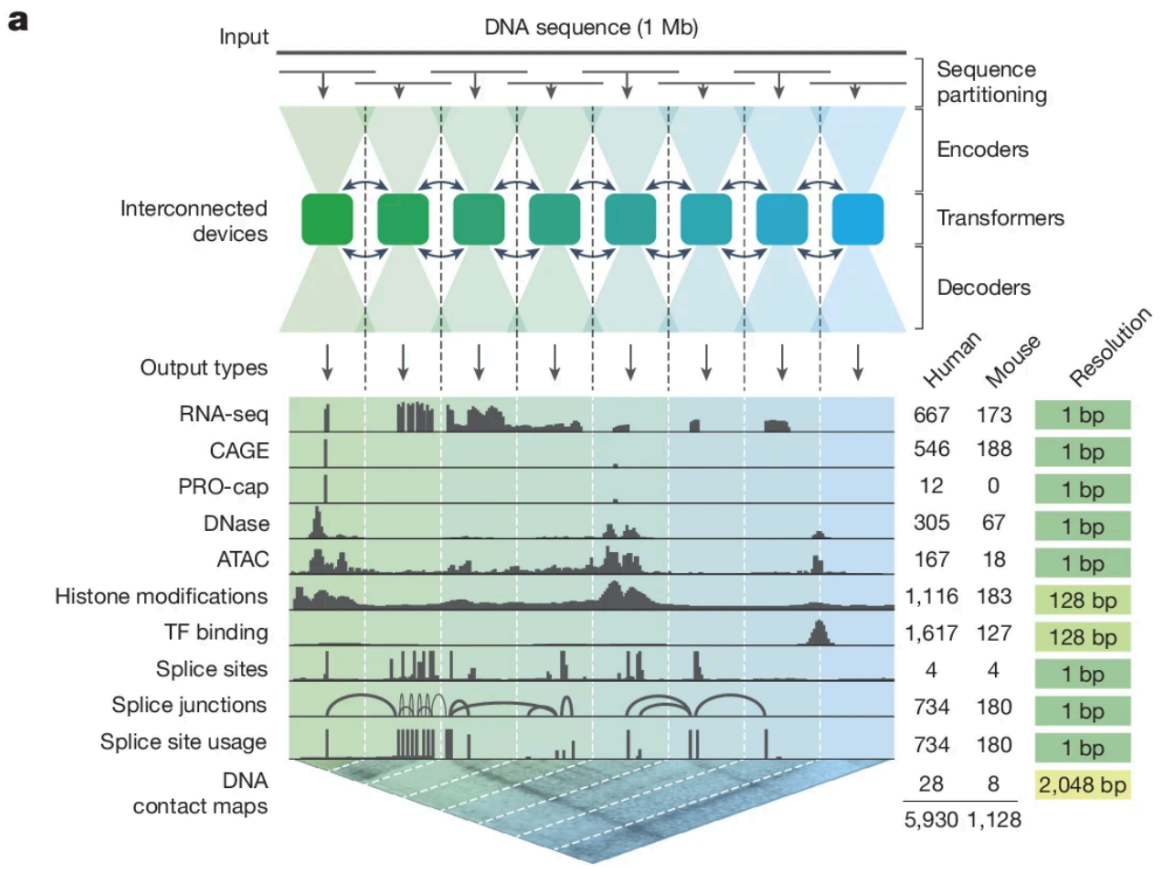
- All methods on the previous slide rely on good LD reference data
- With increasing sample sizes, summary-statistic methods are critical
- **GWAS summary statistics** are widely shared
- **LD reference from the same study is not** — this needs to change

What About Deep Learning?

> "In all, over the range of traits evaluated in this study, CNN performance was competitive to linear models, but **we did not find any case where DL outperformed the linear model by a sizable margin.**"

Deep learning with CNNs is not consistently outperforming simple PRS for complex trait prediction.

DNA Sequencing to Gene Expression



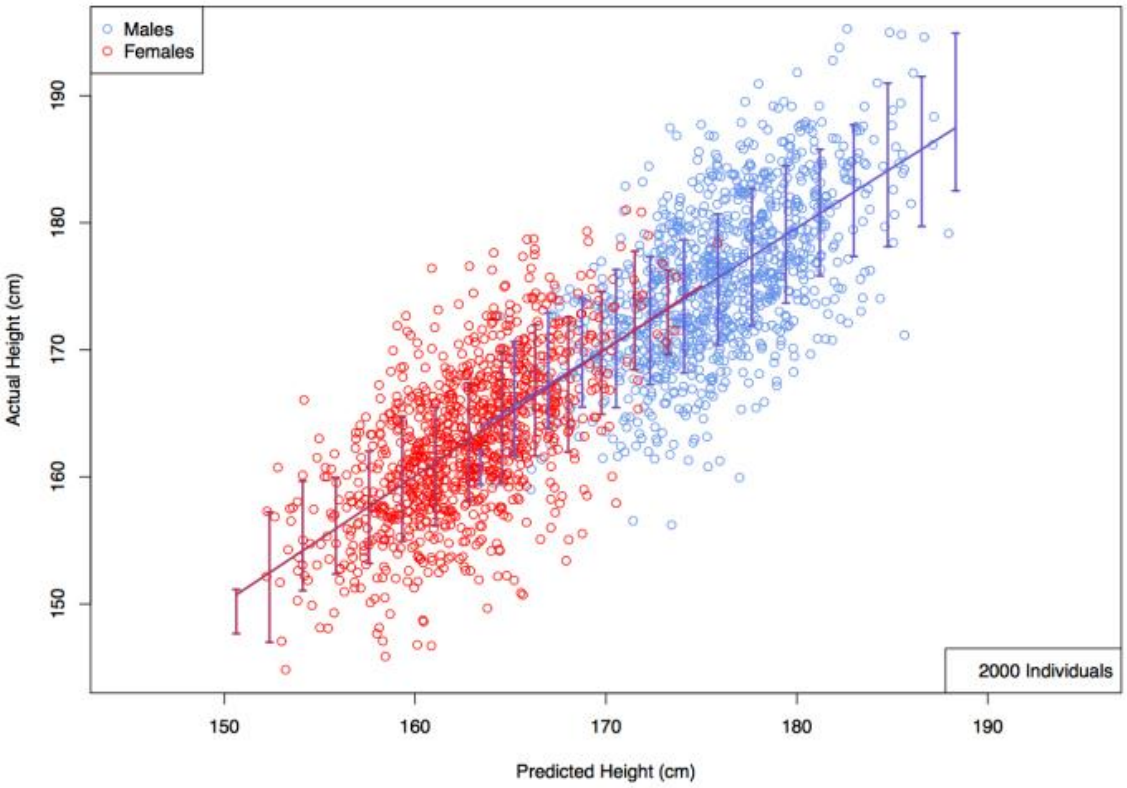
Avsec, Ž., Latysheva, N., Cheng, J. et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature* 649, 1206–1218 (2026).

Transformers + CNN methods (Enformer, from Calico + DeepMind) are promising for predicting mRNA transcription from DNA sequence. This model has 250 million parameters — compare to ChatGPT's 170 billion, highlighting that scale makes a huge difference.

Clinical Utility of Genetic Predictions

An important question is whether PRS predictions have clinical utility.

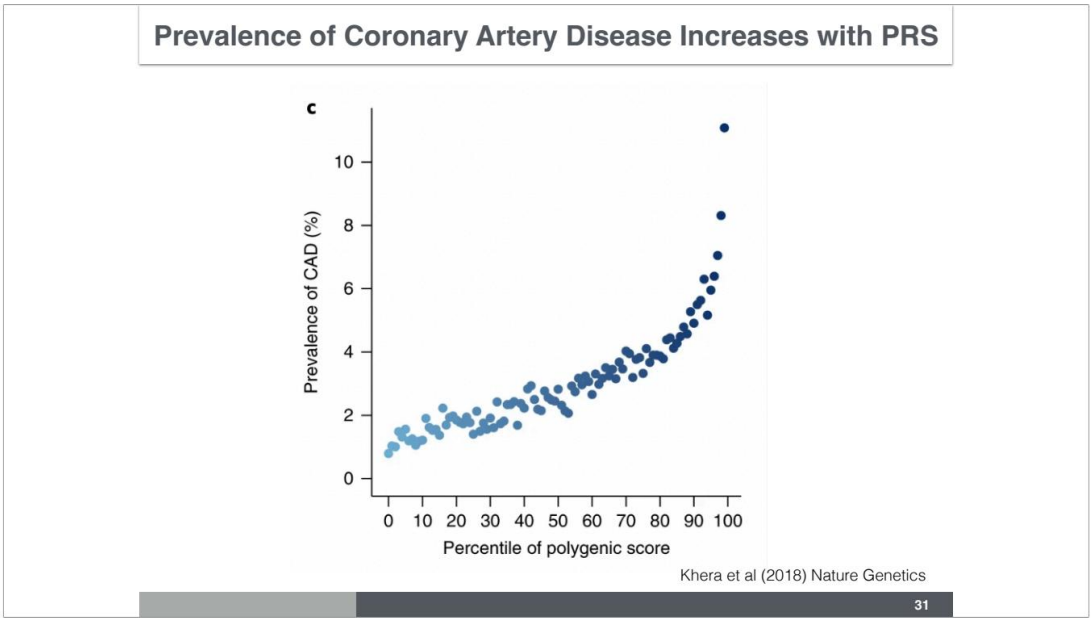
Genomic Prediction of Height in UK Biobank



Lello et al. (2018). Accurate Genomic Prediction of Human Height. *Genetics*

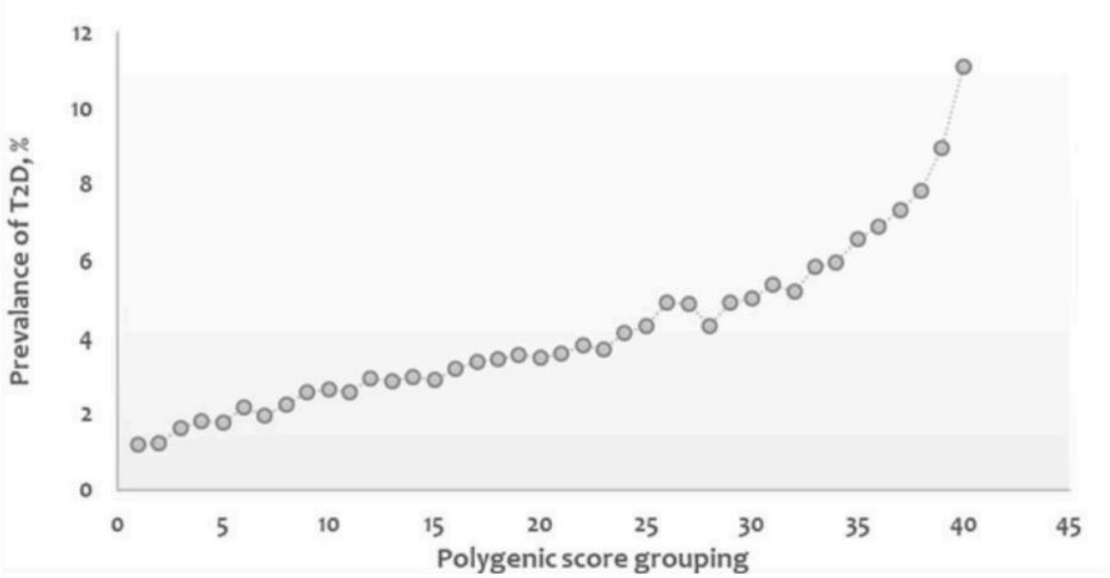
With biobank-scale data, we are able to predict height quite well using common variants.

Prevalence of Coronary Artery Disease Increases with PRS



CAD prevalence increases monotonically with polygenic score percentile. This suggests PRS could be used to stratify individuals into low, medium, and high risk groups for coronary artery disease.

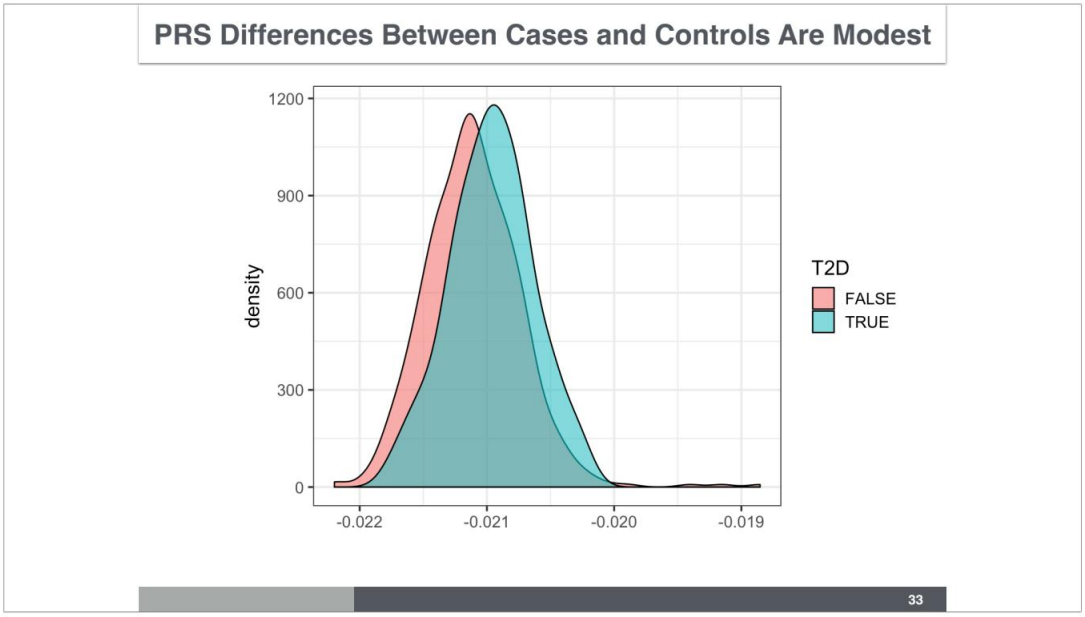
Prevalence of Type 2 Diabetes Increases with PRS



Mahajan et al. (2019). *Nature Genetics*

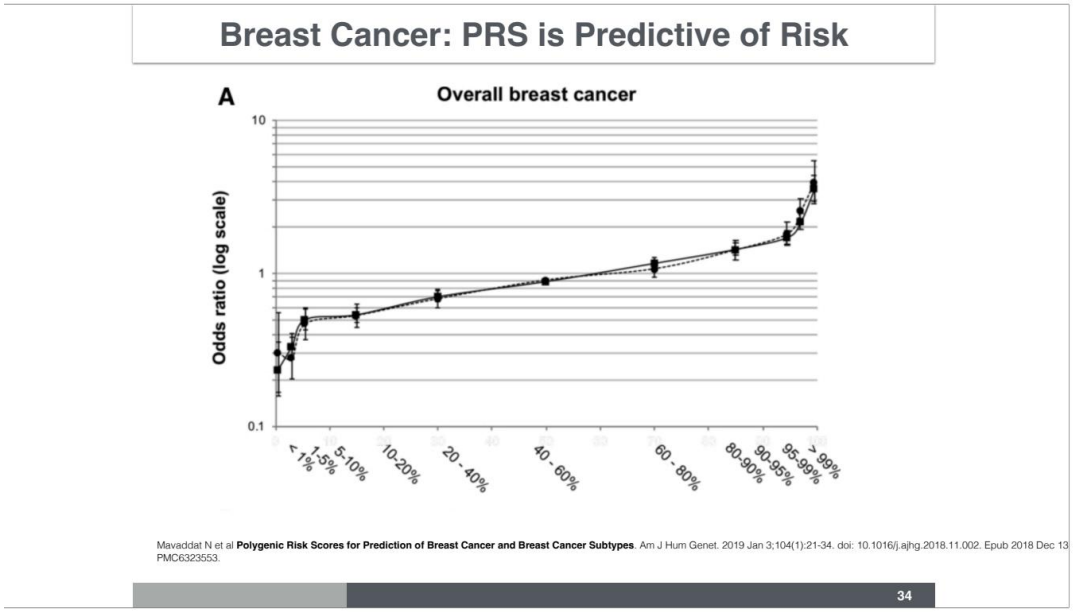
T2D prevalence increases monotonically with polygenic score grouping, consistent with the pattern seen for CAD.

PRS Differences Between Cases and Controls Are Modest



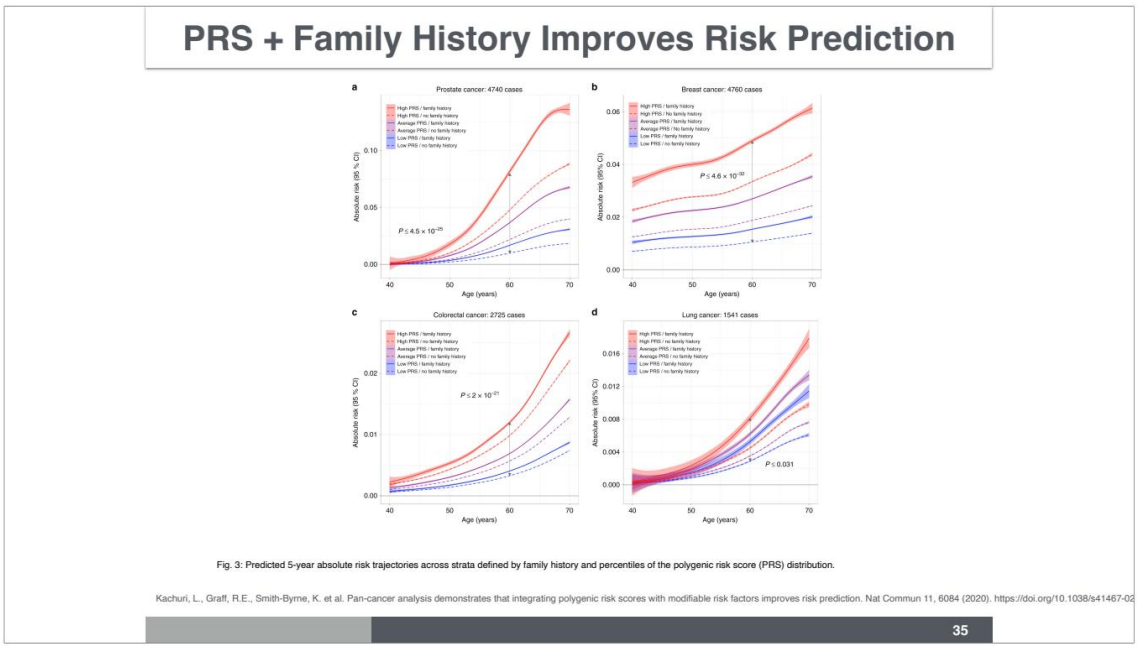
This figure shows the distribution of T2D PRS for cases and controls. There is a clear shift between the two distributions, but they overlap substantially — prediction is probabilistic, not deterministic.

Breast Cancer: PRS Is Predictive of Risk



Individuals in higher percentiles of the breast cancer PRS have higher odds ratios for developing breast cancer. The log-scale y-axis shows the effect is multiplicative.

PRS + Family History Improves Risk Prediction



Predicted 5-year absolute risk trajectories across strata defined by family history and PRS percentile, for prostate, breast, colorectal, and lung cancers. Combining PRS with family history substantially improves risk stratification compared to either alone.

Do PRS Work for Everyone?

Portability of Prediction Across Ancestries

Current GWAS data is mostly collected in **European-descent individuals**

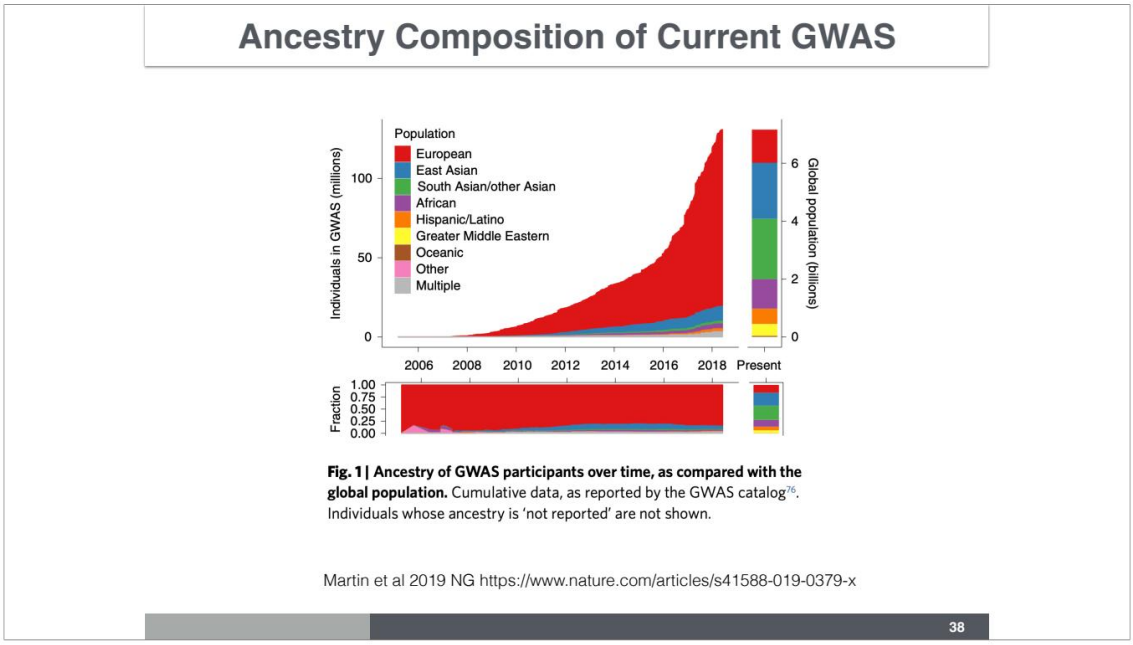
PRS performance deteriorates with genetic distance from EUR training data, due to:

- **Differences in LD patterns** — significant variants are often proxies for causal ones; proxies vary across populations
- **Allele frequency differences** — low-frequency causal variants in non-EUR populations may not be discovered
- **Different causal variants** — not all causal variants are shared across ancestries

Martin et al. (2019). Clinical use of current polygenic risk scores may exacerbate health disparities. *Nature Genetics*

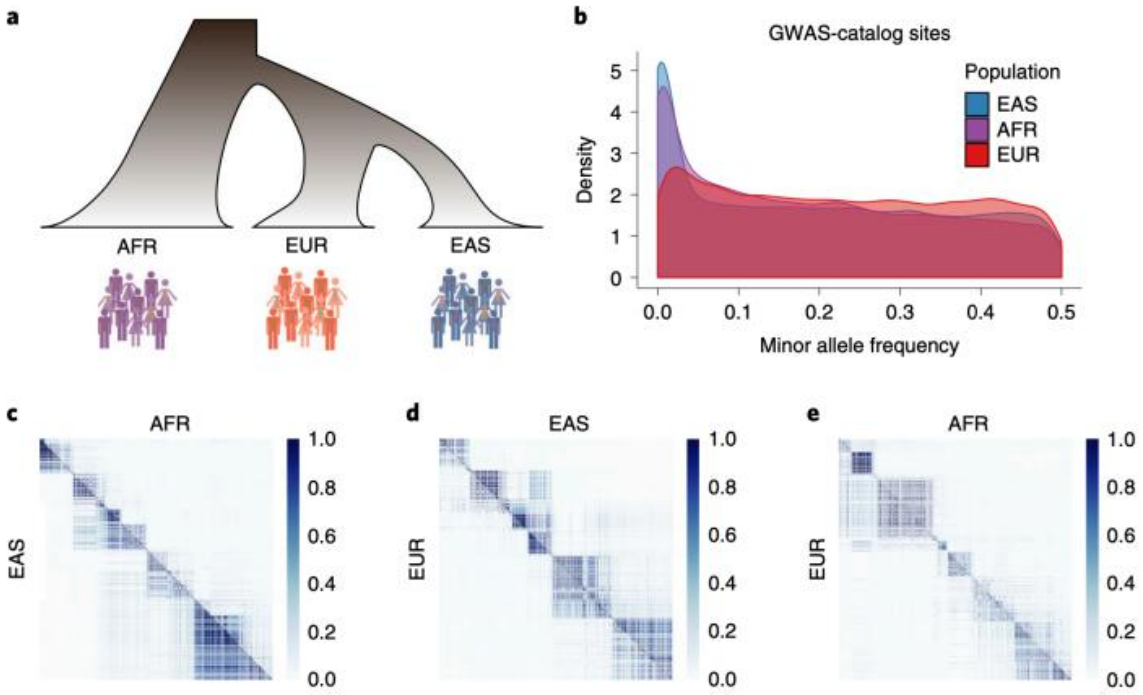
The worry is that because current GWAS data is mostly collected in European-descent individuals, performance of PRS may deteriorate with genetic distance from EUR ancestry. This has important implications for health equity.

Ancestry Composition of Current GWAS



More than 80% of GWAS have been performed on European-descent individuals, who represent less than 15% of the world population.

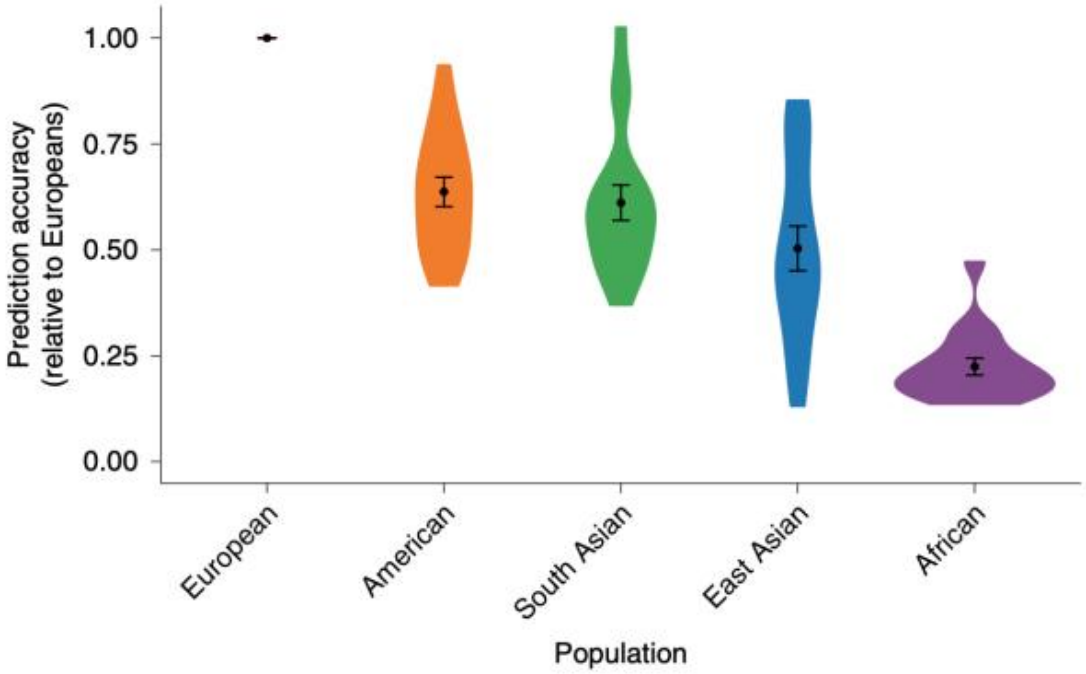
Allele Frequency and LD Differ Across Ancestries



Martin et al. (2019). *Nature Genetics*

AFR, EAS, and EUR populations differ substantially in minor allele frequency distributions at GWAS catalog sites (panel b) and in LD structure (panels c–e). These differences drive the portability problem.

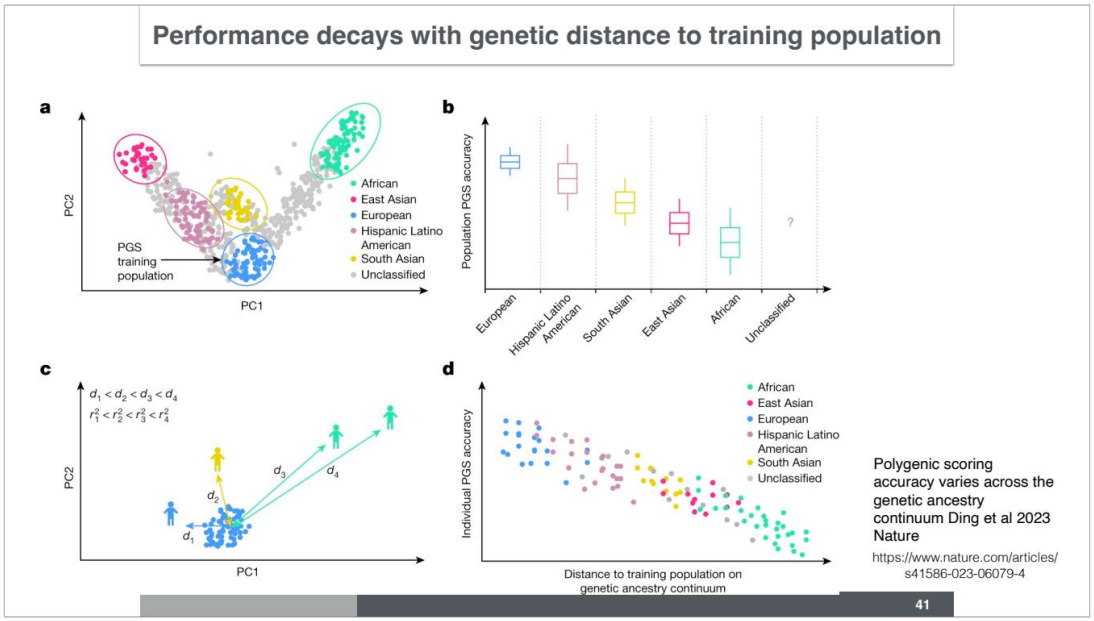
PRS Does Not Transfer Well Across Populations



Martin et al. (2019). *Nature Genetics*

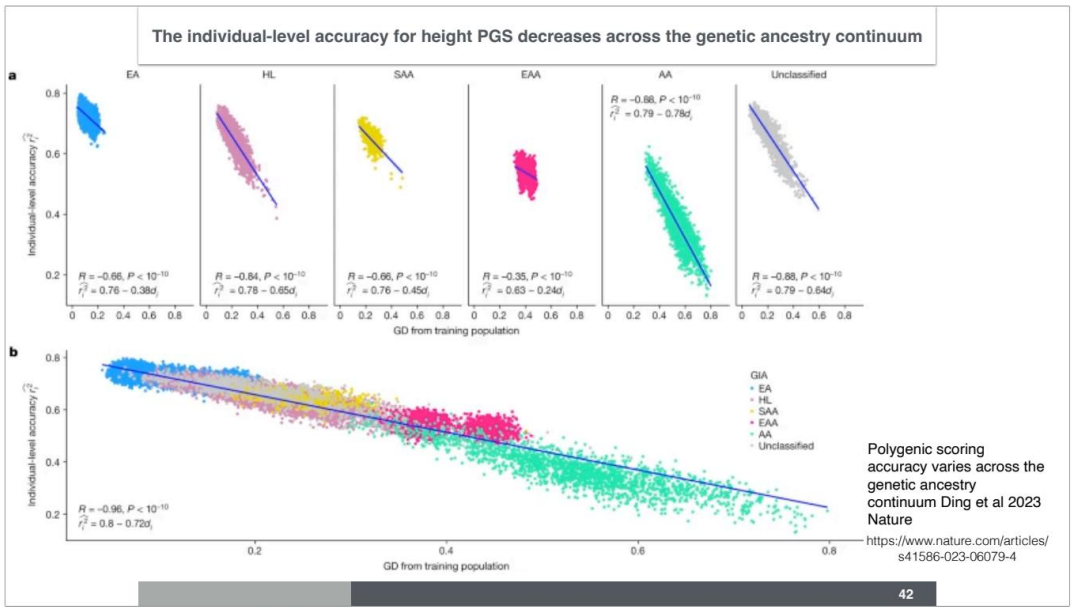
Prediction accuracy decreases as genetic distance from the EUR training population increases. Performance is normalized to EUR (= 1.0 by definition). Among AFR individuals, prediction accuracy drops to ~25% of EUR performance.

Performance Decays with Genetic Distance to Training Population



Individual PGS accuracy decays along the genetic ancestry continuum — both at the population level (discrete GIA clusters) and at the individual level as a continuous function of genetic distance from the training population.

Individual-Level PGS Accuracy Across the Genetic Ancestry Continuum



Individual-level PGS accuracy for height decreases within both homogeneous and admixed ancestry groups. The correlation between genetic distance to the training population and PGS accuracy is highly significant across all groups (R -0.88 to -0.35).

Investments in More Diverse Samples Are Being Made

- Large initiatives underway: All of Us, PAGE, H3Africa, GenomeAsia
- Multi-ancestry GWAS and meta-analyses are increasing
- Methods for multi-ancestry PRS actively being developed
- Goal: PRS that work equitably across all ancestries

Bioethical Issues with Embryo Screening

Screening Human Embryos for Polygenic Traits Has Limited Utility

Key findings:

- PRS offered since 2019 to screen IVF embryos for adult disease risk
- Excluding embryos with **very high** PRS → minimal risk reduction

- Selecting embryo with **lowest** PRS → substantial relative risk reduction, given enough viable embryos
- Utility depends on: variance explained by PRS, number of embryos, disease prevalence, parental PRSs

Karavani et al. (2019). Screening human embryos for polygenic traits has limited utility. *Cell* 179(6): 1424–1435

PRS have been offered since 2019 to screen in vitro fertilization embryos for genetic liability to adult diseases, despite a lack of comprehensive modeling of expected outcomes.

Embryo Selection: Active Area of Research

This eLife paper shows that combining rare variants with PRS magnifies predicted differences across sibling embryos. In a couple with a pathogenic BRCA1 variant, combining BRCA1 + PRS predicted a 15-fold difference in odds ratio across siblings — versus 4.5-fold for BRCA1 alone or 3-fold for PRS alone.

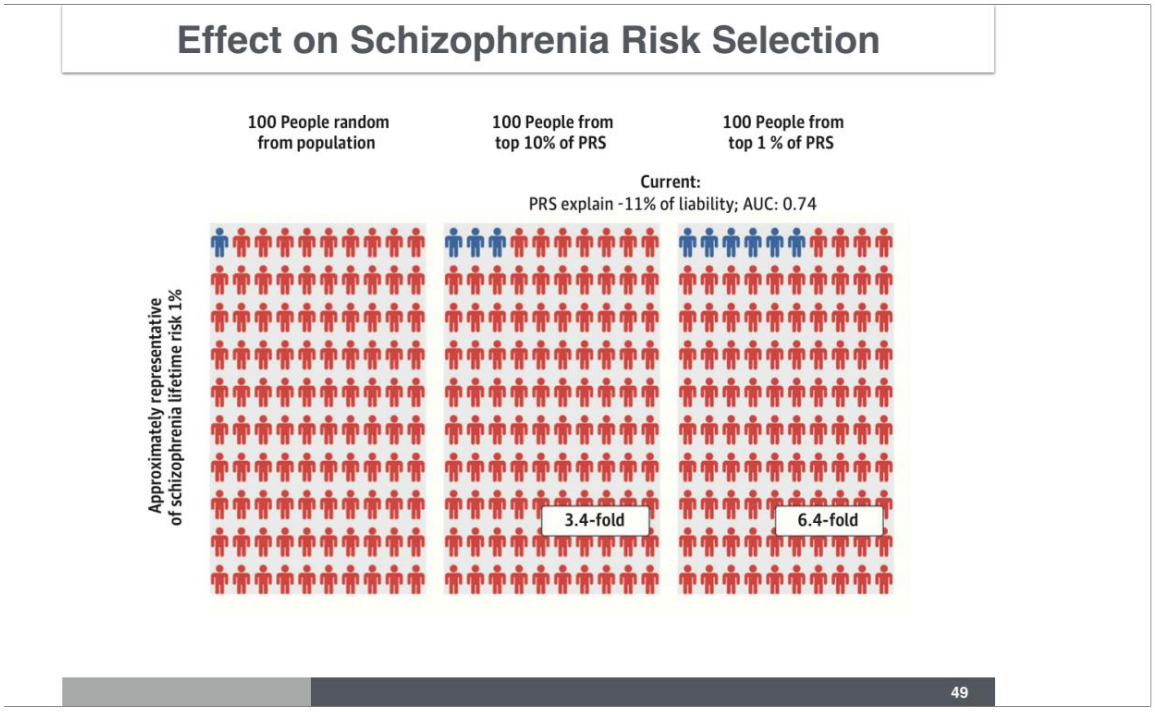
Orchid Offers Preconception Testing

Are they helping **eliminate disease** — or implementing **eugenics**?

- Preimplantation PRS screening is **illegal in Europe**; sold commercially in the US
- Lior Pachter's critique: "The amoral nonsense of Orchid's embryo selection"

Preimplantation screening with PRS is illegal in Europe but offered commercially in the US by companies like Orchid. The European Society of Human Genetics has issued a statement against preimplantation screening based on PRS.

Effect on Schizophrenia Risk Selection



PRS prediction is probabilistic. This figure shows the proportion of 100 people who may develop schizophrenia: if randomly selected from the population (left), if selected from the top 10% of PRS (middle, 1.4-fold enrichment), and if selected from the top 1% of PRS (right, 5.4-fold enrichment). Current PRS explains ~11% of liability (AUC = 0.74).

Key Takeaways

- **PRS** estimate genetic susceptibility to cancers and other complex diseases by summing GWAS effect sizes across the genome
- Several statistical approaches address the too-many-parameters problem: random effects, one-SNP-at-a-time GWAS, and penalized regression (ridge, LASSO, elastic net)
- PRS have potential clinical utility in risk stratification, but predictive accuracy is **modest** — distributions of cases and controls overlap substantially
- PRS accuracy degrades with genetic distance from the training population (mostly European) — **ancestry bias** is a major limitation
- Embryo screening using PRS raises serious **ethical concerns** and is restricted or illegal in many jurisdictions